

AF model parameters × behavioral RT contrasts

Model parameters per (subject, ROI):

$\log_ratio = \log((1+g_HP)/(1+g_LP))$

$g_avg = \frac{1}{2}(g_HP + g_LP)$

σ_AF

Behavioral contrasts per subject:

$RT_HP - RT_LP$ (HP-vs-LP cost; usually NEGATIVE)

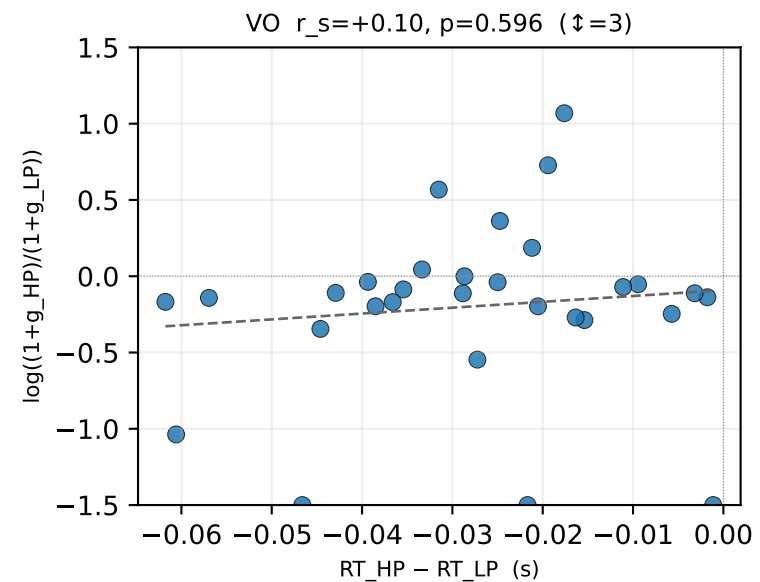
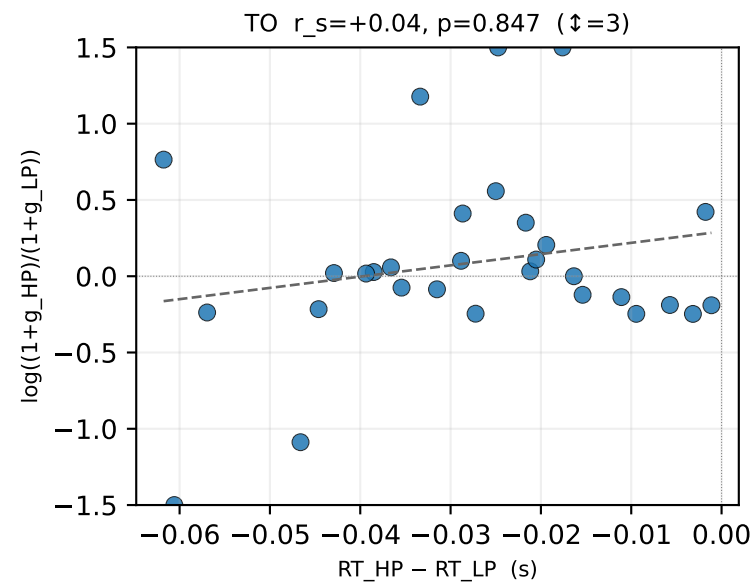
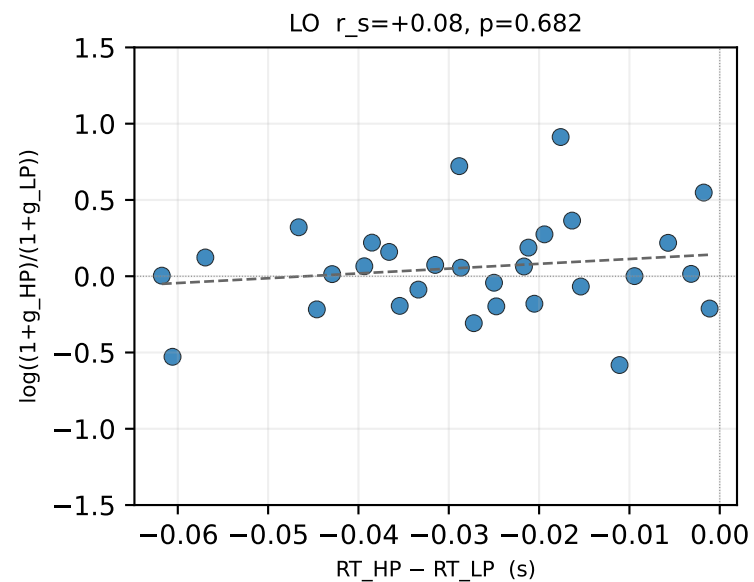
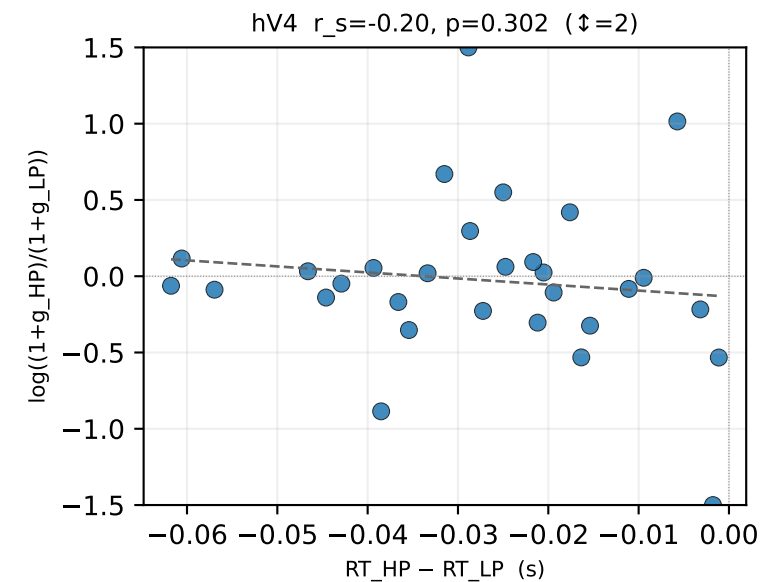
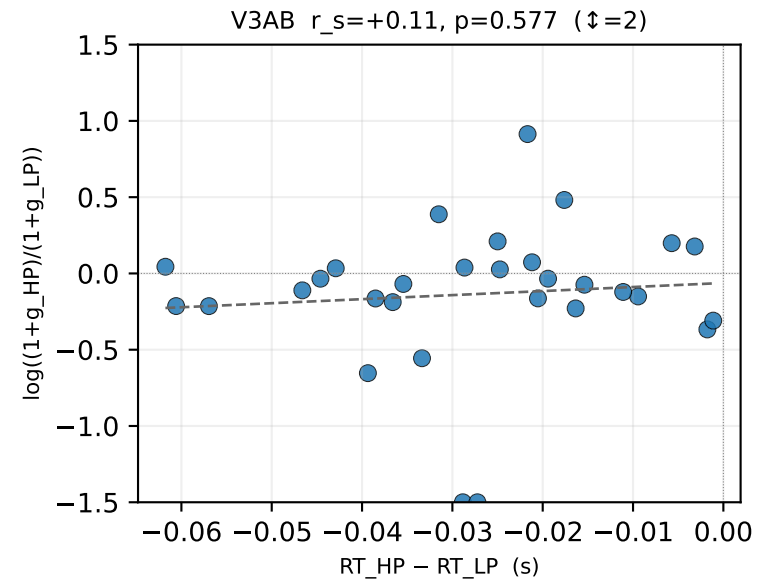
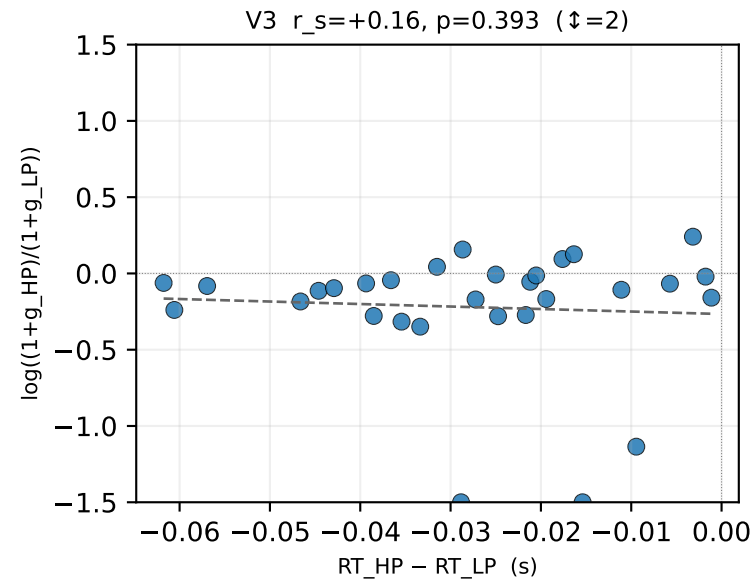
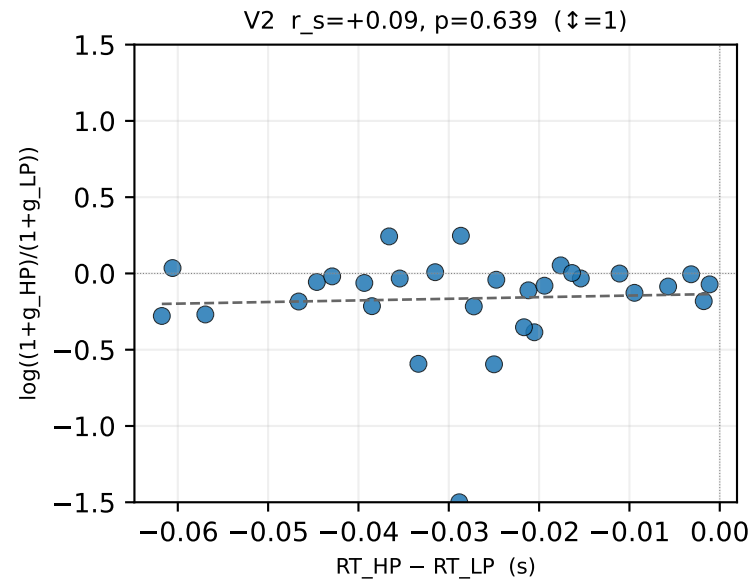
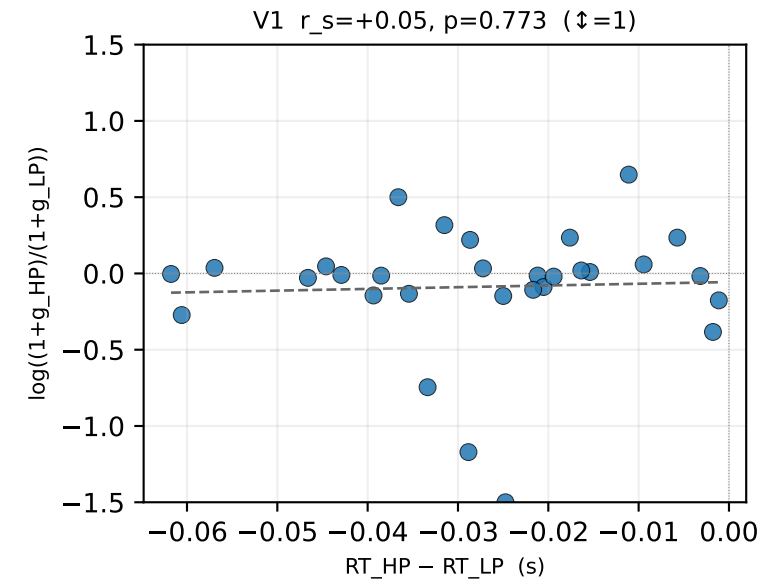
$RT_dist - RT_no_dist$ (any distractor cost; positive)

All correlations are SPEARMAN (rank-based, robust to fit outliers).

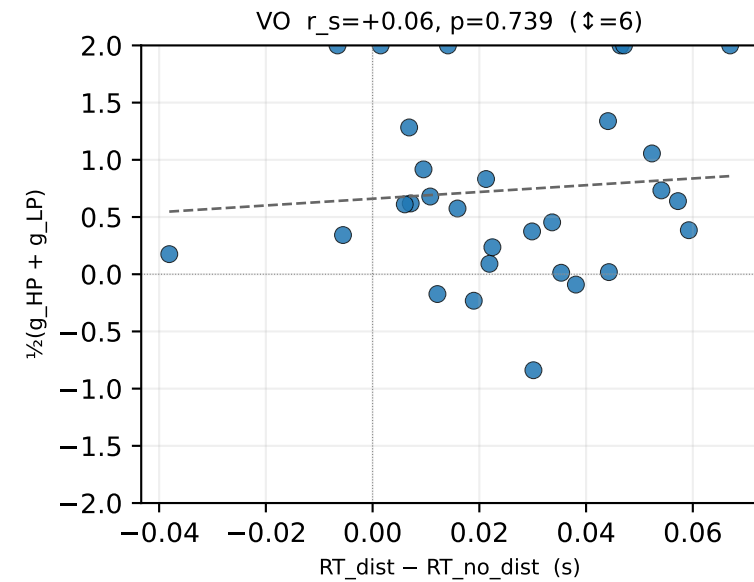
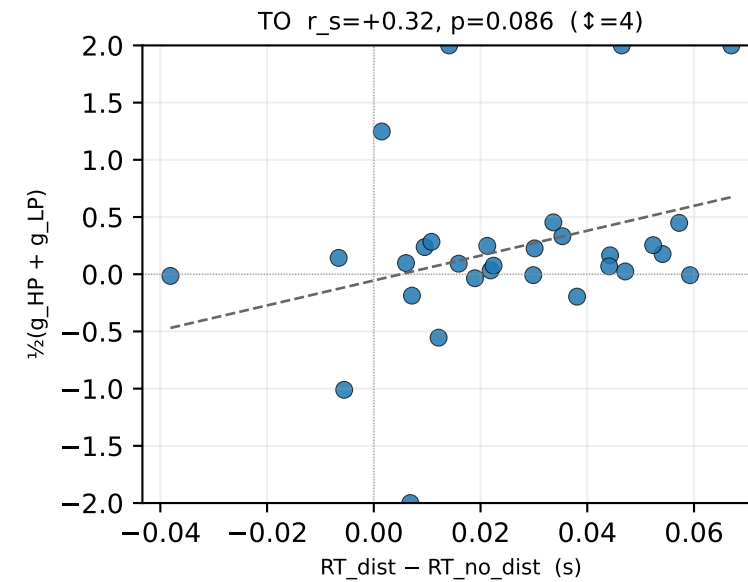
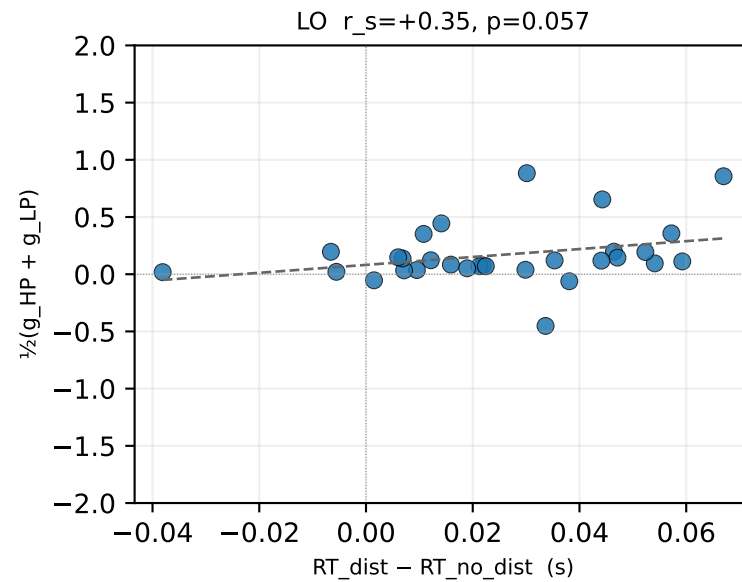
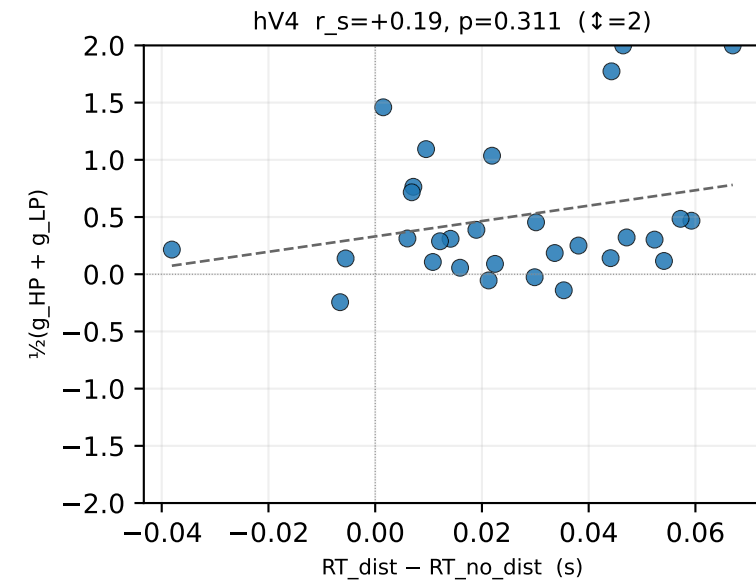
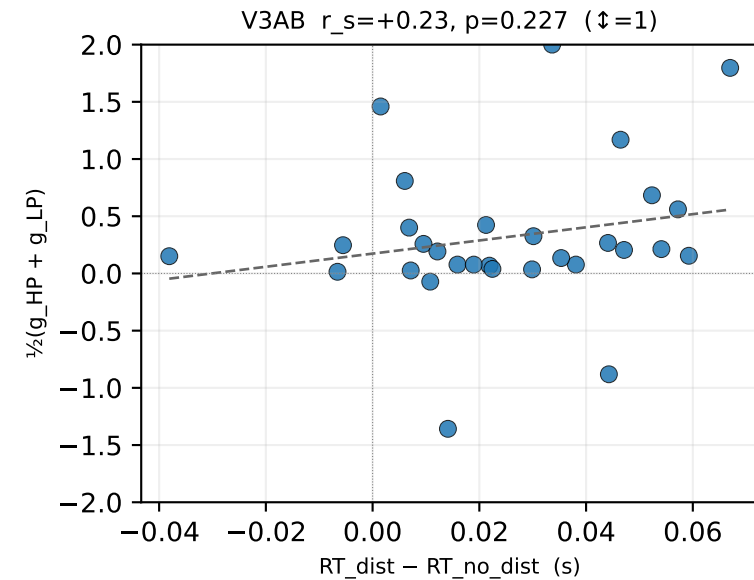
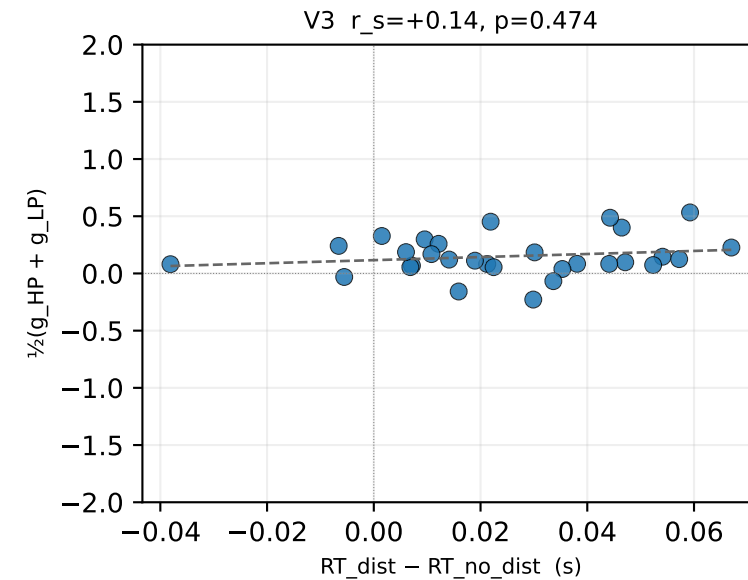
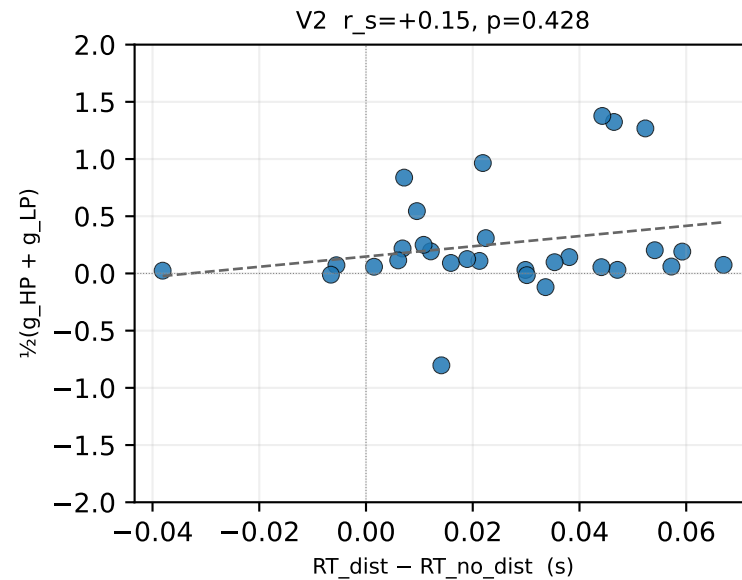
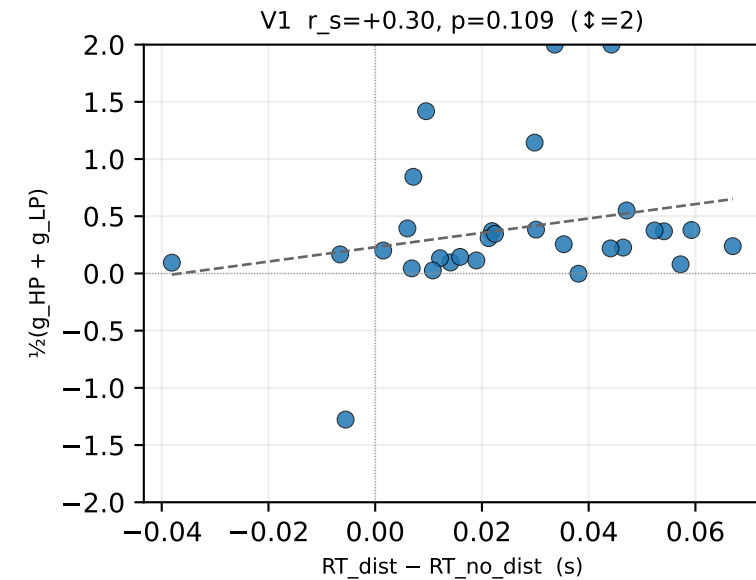
Expected directions if the model is recovering real attention:

- $\log_ratio$ vs $RT_HP-LP \leftrightarrow$ POSITIVE (both negative)
- g_avg vs $RT_dist-no_dist \leftrightarrow$ POSITIVE
(more attentive engagement $\Rightarrow$ stronger AF gain $\Rightarrow$ more cost)
- σ_AF vs $RT_dist-no_dist \leftrightarrow$ either sign – narrower AF could mean stronger but more localized engagement.

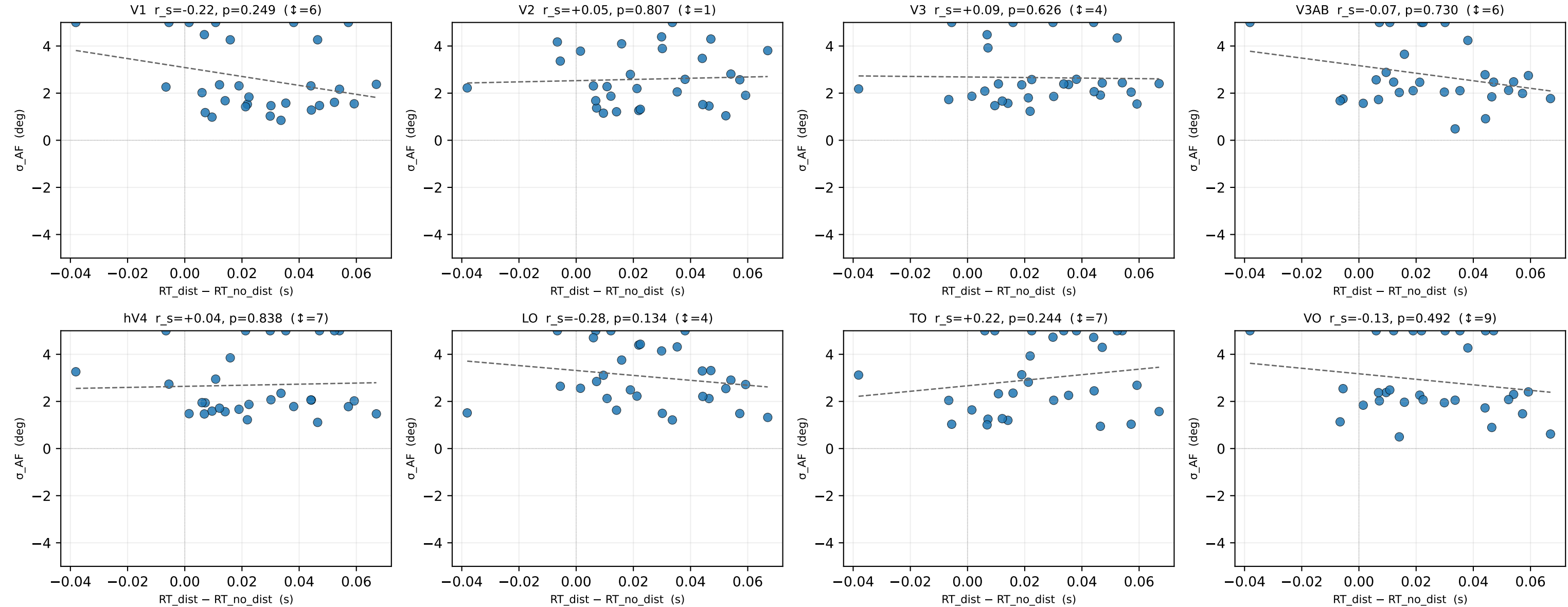
$\log((1+g_{HP})/(1+g_{LP}))$ vs $RT_{HP} - RT_{LP}$ (s)



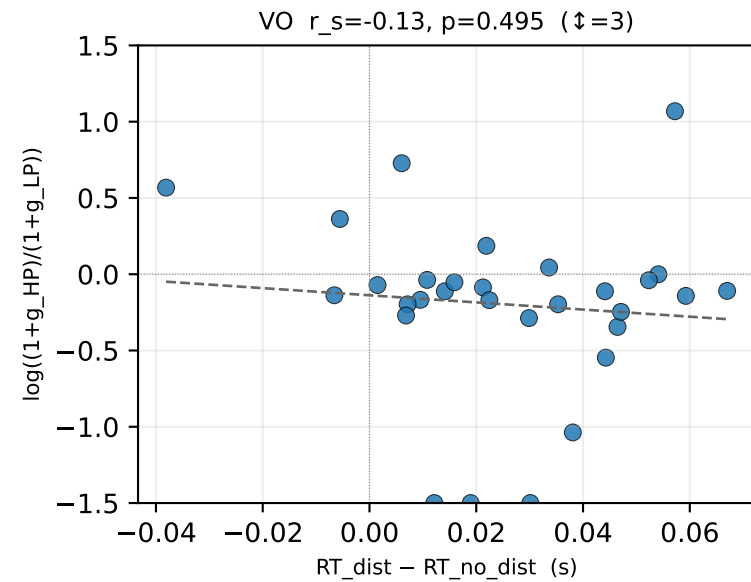
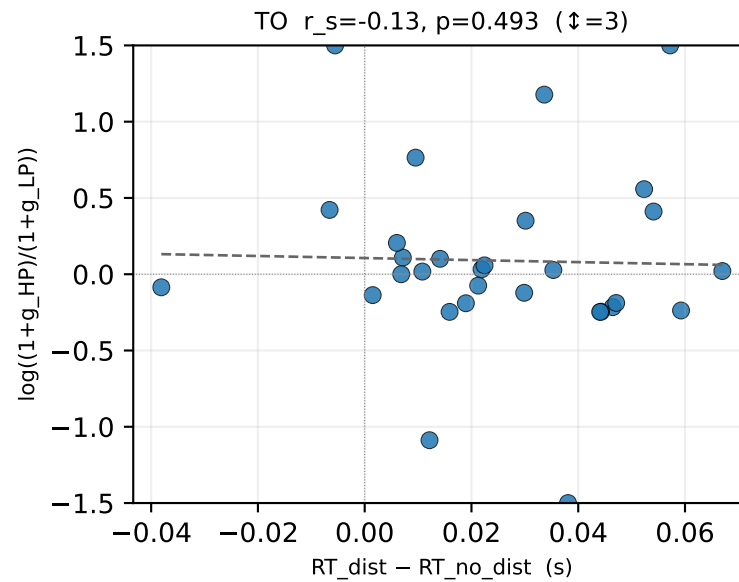
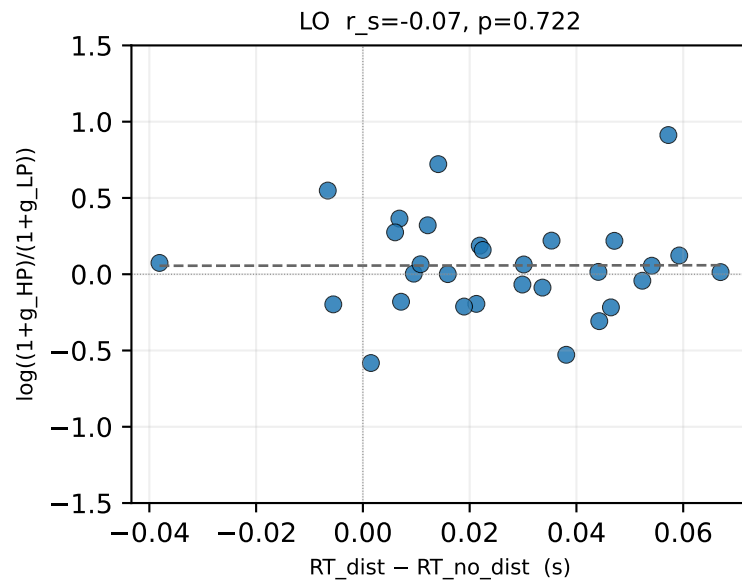
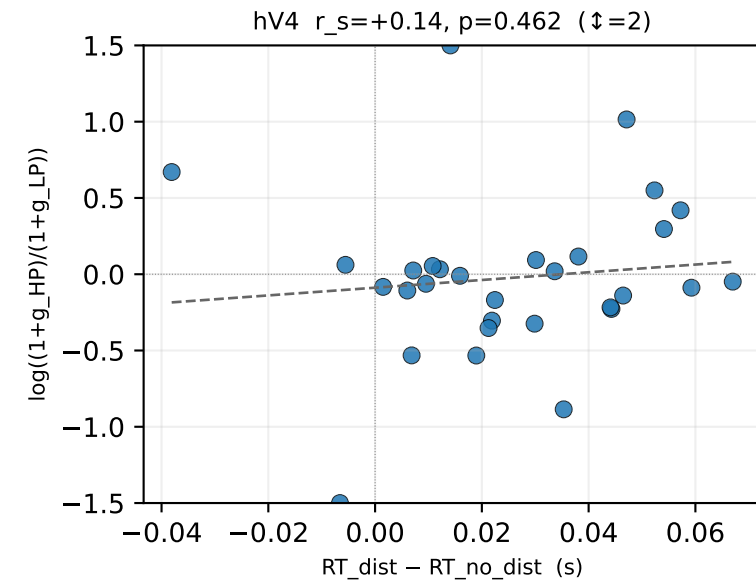
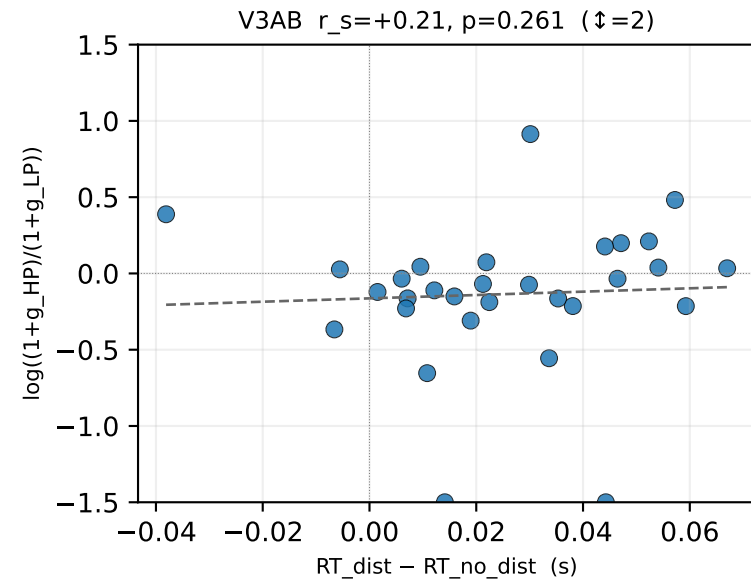
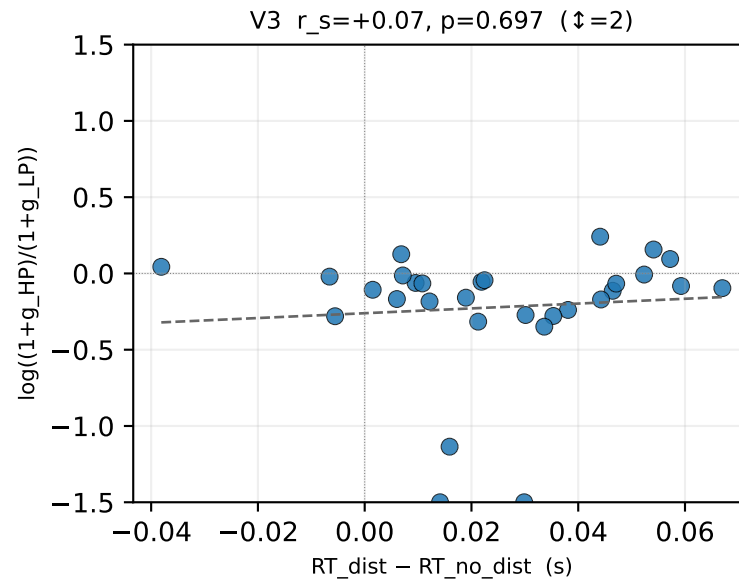
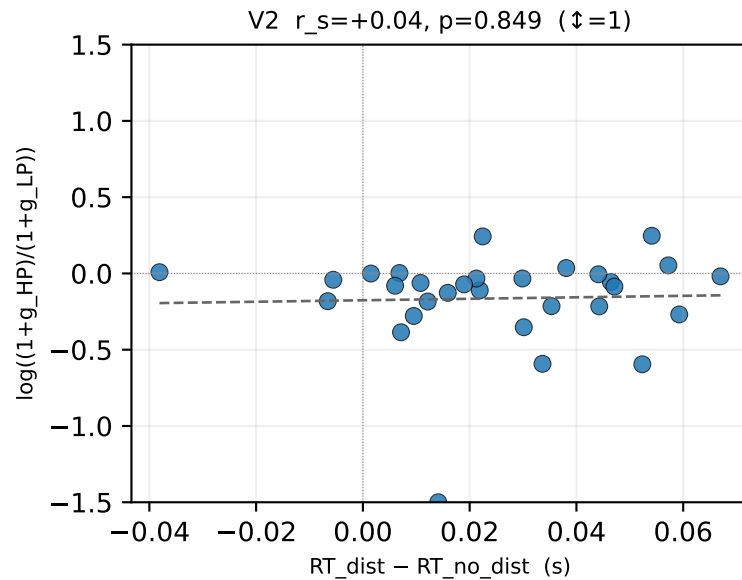
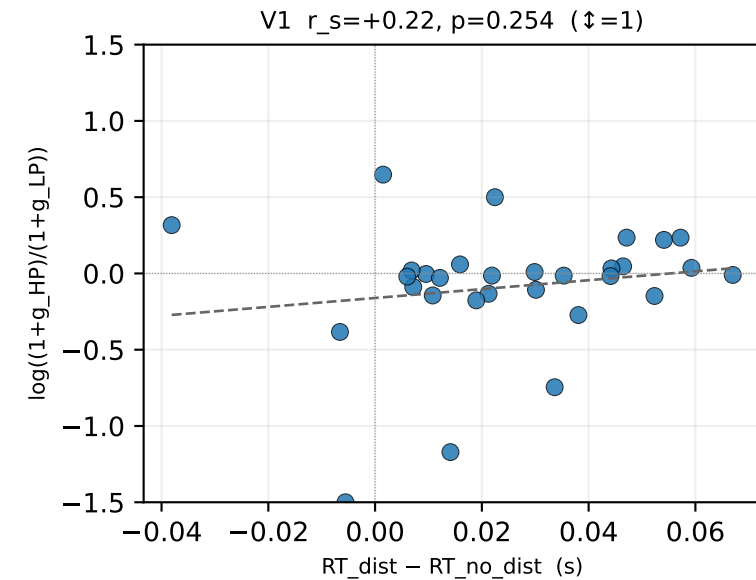
$\frac{1}{2}(g_{HP} + g_{LP})$ vs $RT_{dist} - RT_{no_dist}$ (s)



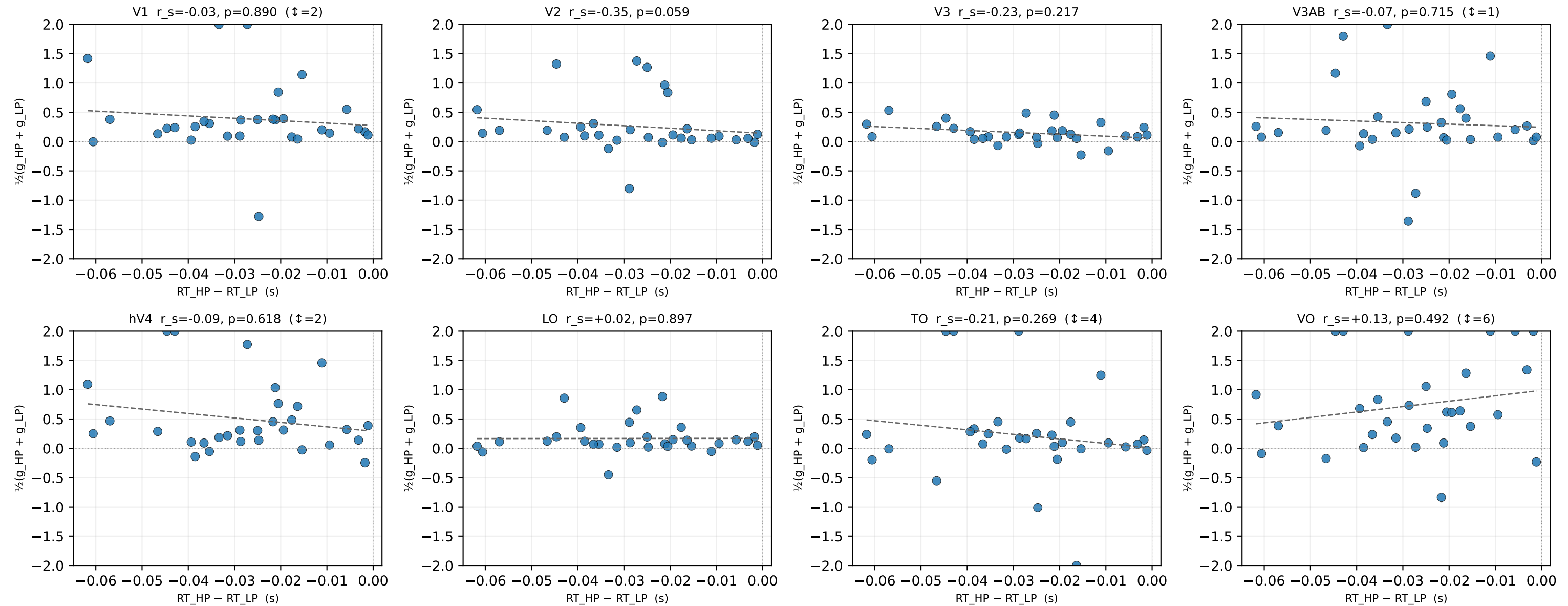
σ_{AF} (deg) vs $RT_{dist} - RT_{no_dist}$ (s)



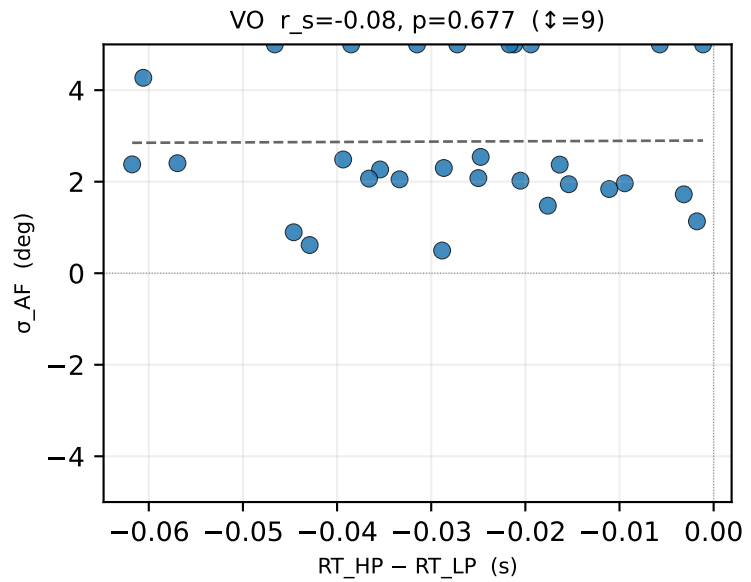
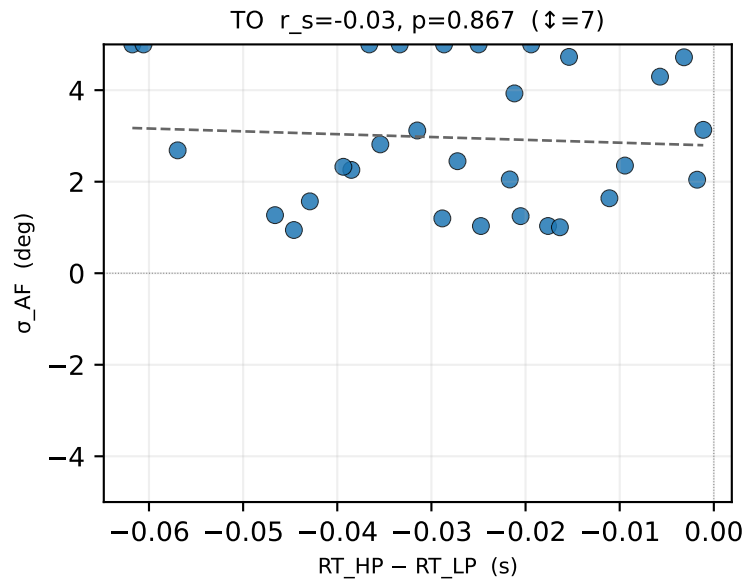
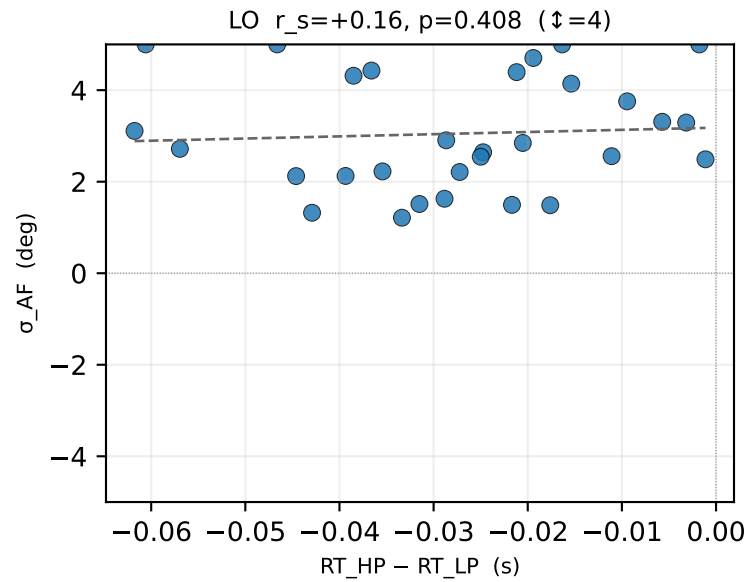
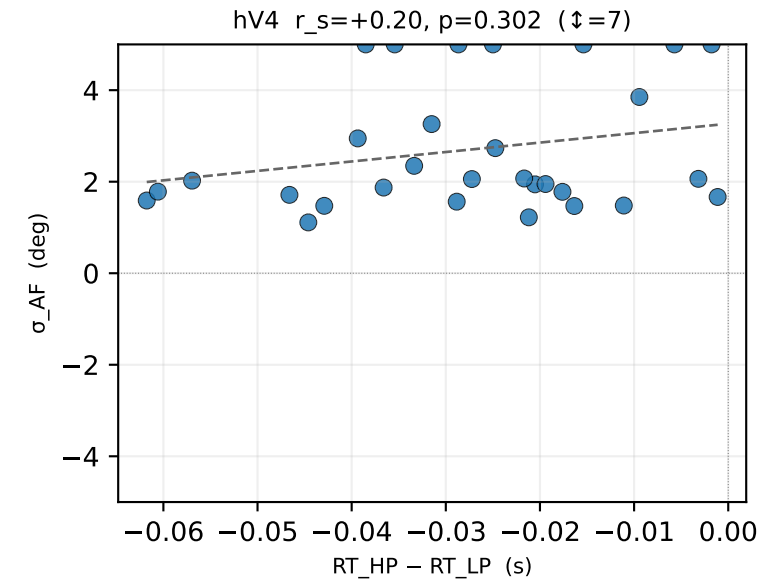
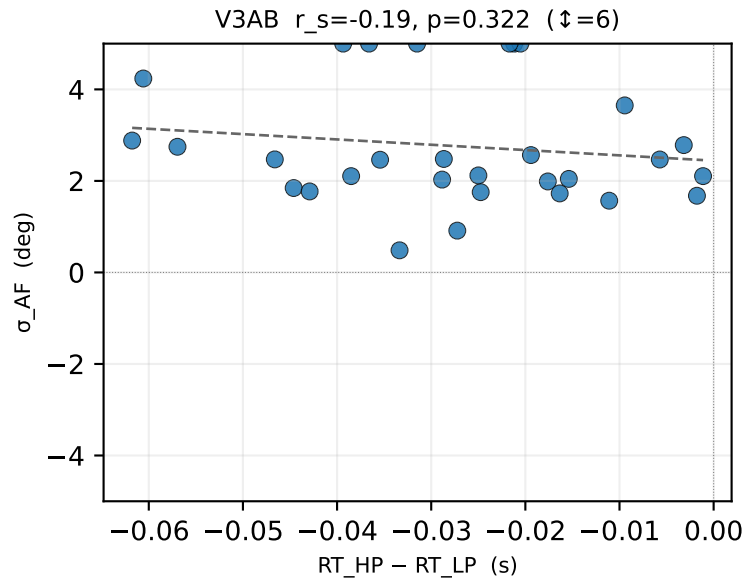
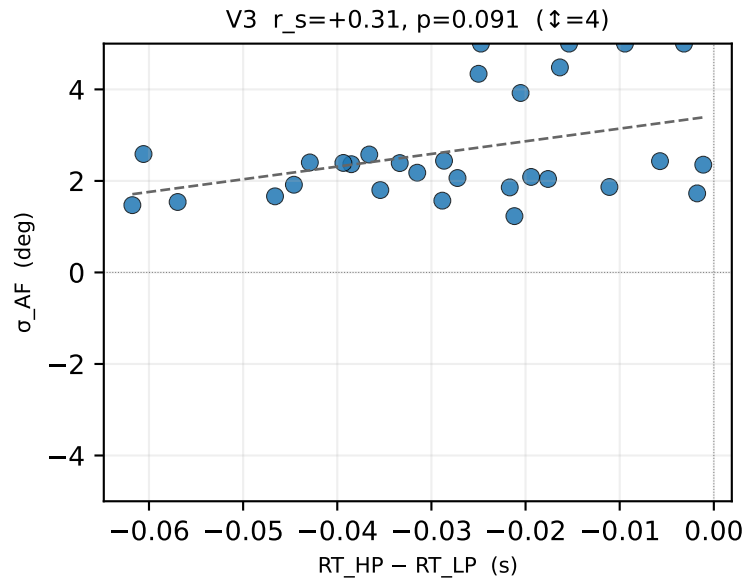
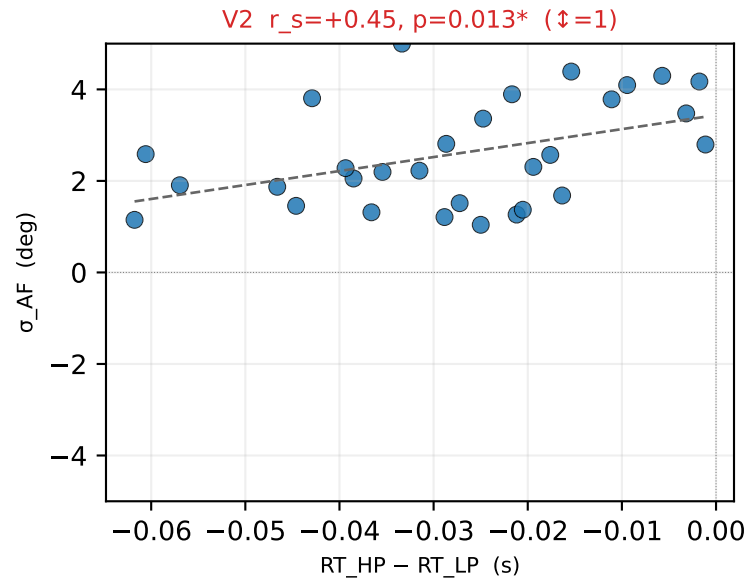
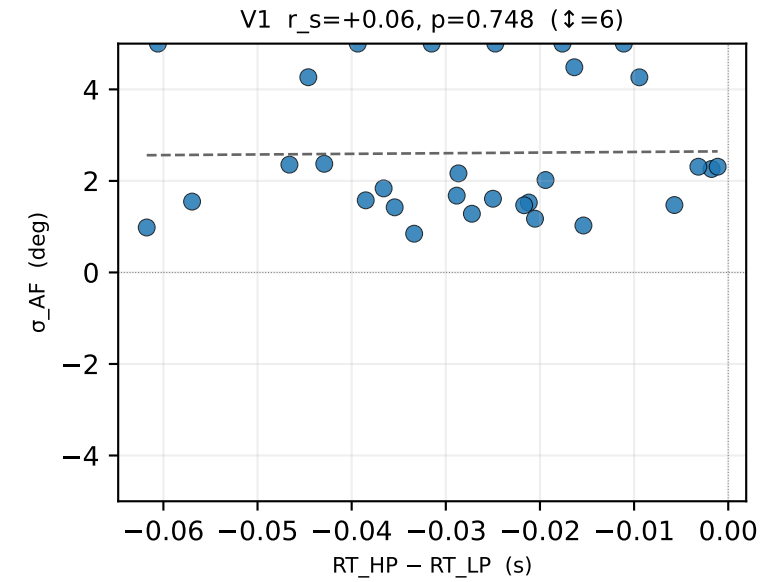
$\log((1+g_{HP})/(1+g_{LP}))$ vs $RT_{dist} - RT_{no_dist}$ (s)



$\frac{1}{2}(g_{HP} + g_{LP})$ vs $RT_{HP} - RT_{LP}$ (s)



σ_{AF} (deg) vs $RT_{HP} - RT_{LP}$ (s)



Per-ROI Spearman r summary across all model × behavior contrasts

	$\log((1+g_{HP})/(1+g_{LP})) \times RT_{HP} - RT_{LP}$	(s) r	$\log((1+g_{HP})/(1+g_{LP})) \times RT_{HP} - RT_{LP}$	(s) p	$\frac{1}{2}(g_{HP} + g_{LP}) \times RT_{dist} - RT_{HP}$
V1		+0.05		0.773	
V2		+0.09		0.639	
V3		+0.16		0.393	
V3AB		+0.11		0.577	
hV4		-0.20		0.302	
L0		+0.08		0.682	
T0		+0.04		0.847	
V0		+0.10		0.596	